

OPERATOR'S MANUAL

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CHAPTER ONE

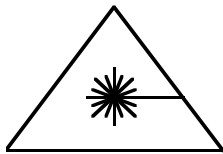
LASER SAFETY

Introduction

The Infinity laser is a high performance laser and has undergone extensive testing to ensure that, with proper usage, it is a safe and reliable system. Due to the nature of laser light and the performance level of the Infinity laser there are some important safety considerations that the you must be aware of. These relate to optical safety, electrical safety, and for the Infinity system, chemical safety. The chemical safety issues are due to CFC 113 used in the cell in which the phase conjugate return beam is generated. In order to use the Infinity system in a safe way, it is important that all users, and everyone near the laser system, are aware of the dangers involved.

Optical Safety

Laser light, because of its special properties, poses safety hazards not associated with light from conventional sources. The safe use of lasers requires that all laser users, and everyone near the laser system, are aware of the dangers involved. The safe use of the laser depends upon the user being familiar with the instrument and the properties of coherent, intense beams of light.



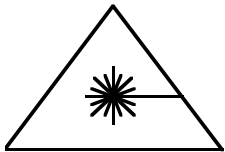
Direct eye contact with the output beam from the laser will cause serious damage and possible blindness.

The greatest concern when using a laser is eye safety. In addition to the main beam, there are often many smaller beams present at various angles near the laser system. These beams are formed by specular reflections of the main beam at polished surfaces such as lenses or beamsplitters. While weaker than the main beam, such beams may still be sufficiently intense to cause eye damage.

Laser beams are powerful enough to burn skin, clothing or paint. They can ignite volatile substances such as alcohol, gasoline, ether and other solvents, and can damage light-sensitive elements in video cameras, photomultipliers and photodiodes. The laser beam can ignite substances in its path, even at some distance. The beam may also cause damage if contacted indirectly from reflective surfaces. For these reasons, and others, the user is advised to follow the precautions below.

1. Observe all safety precautions in the preinstallation and operator's manual.

2. Extreme caution should be exercised when using solvents in the area of the laser.
3. Limit access to the laser to qualified users who are familiar with laser safety practices and who are aware of the dangers involved.
4. Never look directly into the laser light source or at scattered laser light from any reflective surface. Never sight down the beam into the source.
5. Maintain experimental setups at low heights to prevent inadvertent beam-eye encounter at eye level.

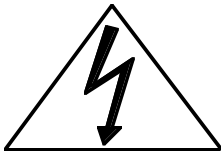


Laser safety glasses can present a hazard as well as a benefit; while they protect the eye from potentially damaging exposure, they block light at the laser wavelengths, which prevents the operator from seeing the beam. Therefore, use extreme caution even when using safety glasses.

6. As a precaution against accidental exposure to the output beam or its reflection, those using the system should wear laser safety glasses as required by the wavelength being generated.
7. Avoid direct exposure to the laser light. The intensity of the beam can easily cause flesh burns or ignite clothing.
8. Use the laser in an enclosed room. Laser light will remain collimated over long distances and therefore presents a potential hazard if not confined.
9. Post warning signs in the area of the laser beam to alert those present.
10. Advise all those using the laser of these precautions. It is good practice to operate the laser in a room with controlled and restricted access.

Electrical Safety

Lethal voltages and currents are present in both the Infinity power supply and laser head. For every laser pulse, up to 50 Joules of energy is deposited into the flash lamp, and the voltage across the capacitor bank can be as high as 570 V. High voltages are also present at various points on the circuit boards. These units are therefore designed to be operated with the protective covers in place. Certain procedures such as changing the flash lamp, water filter, or cleaning optical components, require removal of the protective covers. It is important that all safety precautions contained in this manual are observed by anyone using the laser. The most important rule when performing work on any part of the laser is first to switch it off completely!



There are no user serviceable parts in the electrical side of the power supply. Service procedures on system electronics must be carried out by a Coherent certified field service engineer. Always switch off the laser when working on any part of it.

Troubleshooting and adjustments on the circuit boards are strictly service procedures and should only be done by a Coherent field service technician. Unauthorized attempts by the user to modify or repair the circuit boards will void any warranty!

Chemical Safety

The Infinity laser incorporates a cell filled with CFC 113, which is used to generate the backward traveling return beam going through the power amplifiers. The choice of CFC 113 is motivated by its superior physical properties with respect to reflectivity, chemical stability and threshold for the phase conjugation process. Under normal operating conditions there are no situations in which the user will be exposed to the liquid. In the event of a cell leak or breakage, avoid direct skin and eye contact with the liquid. Ensure good ventilation until it has evaporated completely and the environment is odor free (CFC 113 has an ethereal and a faint sweetish odor). Call Coherent technical support for information about replacement cells and handling of damaged cells.

Safety Features and Compliance to Government Requirements

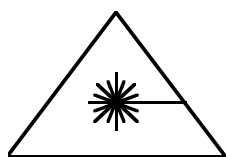
The following features are incorporated into the instrument to conform to several government requirements. The applicable United States Government requirements are contained in 21 CFR, subchapter J, part II administered by the Center for Devices and Radiological Health (CDRH). The European Community requirements for product safety are specified in the "Low Voltage Directive" (published in 73/23/EEC and ammended in 93/68/EEC). The "Low Voltage Directive" requires that lasers comply with the standard EN 61010-1 "Safety Requirements For Electrical Equipment For Measurement, Control and Laboratory Use" and EN60825-1 "Radiation Safety of Laser Products". Compliance of this laser with the requirements is certified by the CE mark.

Classification

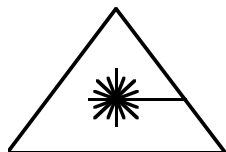
The governmental standards and requirements specify that the laser must be classified according to the output power or energy and the laser wavelength. The Infinity is classified as Class IV based on 21 CFR, subchapter J, part II, section 1040-10 (d). According to the European Community standards, the Infinity classified as Class 4 based on EN 60825-1, clause 9. In this manual and other documentation of the Infinity, the classification will be referred to as Class 4.

Protective Housing

The Infinity head is enclosed in a protective housing that prevents human access to radiation in excess of the limits of Class I radiation as specified in the Federal Register, July 31, 1975, Part II, Section 1040.10 (f) (1) and Table 1-A / EN 60825-1, clause 4.2 except for the output beam, which is Class IV.



Operation of laser with cover removed will allow access to hazardous visible and invisible radiation. The laser housing should only be opened for the purposes of maintenance and service by trained personnel cognizant of the hazards involved.



Use of controls or adjustments or performance of procedures other than those specified in the manual may result in hazardous radiation exposure.



Use of the system in a manner other than that described herein may impair the protection provided by the system.

Safety Interlocks

Electrical safety interlock breakers are positioned at four places in the Infinity system to prevent accidental exposure to dangerous levels of electricity or laser radiation. (21 CFR 1040.10 (f)(2)). The four interlocks are: head cover (1), flash lamp terminal housing (1) and amplifier housing latches (2).

External Interlock Connector

A RCA plug on the rear of the power supply is provided for use as an external interlock connection. This connector must be short circuited for the laser to operate. The laser is shipped with a plug installed. If this is removed then an external interlock circuit can be connected in its place (21 CFR 1040.10 (f)(3)).

Key Control

The instrument cannot be turned on or operated until the system key switch, located at the bottom of the power supply beneath the BNC connectors, is turned on (21 CFR 1040.10 (f)(4)).

Laser Radiation Emission Indicator

The appropriately labeled indicator lights on the laser head are turned on at least five seconds before laser emission can occur. It is visible without exposing the operator to laser emission. White light is used so that it may be seen regardless of the type of safety glasses employed (21 CFR 1040.10 (f)(5)).

Beam Attenuators

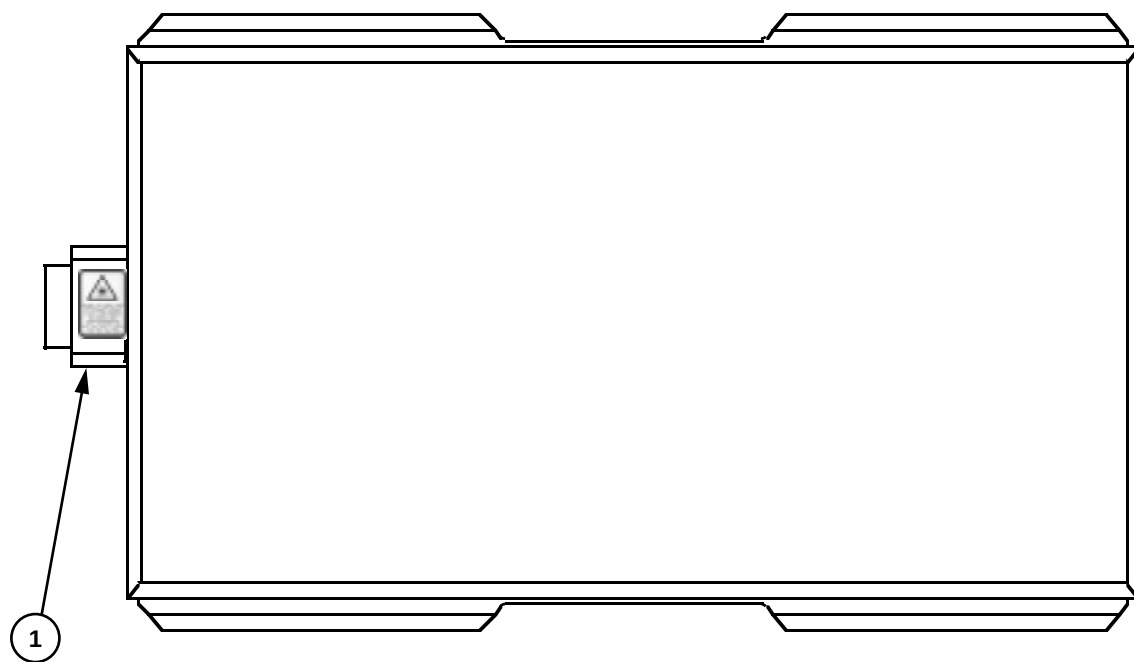
A mechanical shutter is provided with the Infinity. The shutter is actuated through the computer interface (21 CFR 1040.10 (f)(6)).

Manual Reset

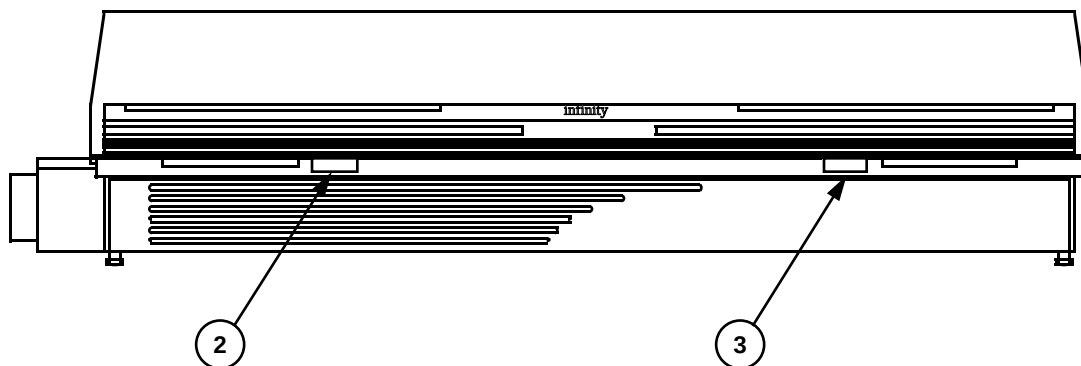
After an interruption in laser activity because of remote interlock activation, or from an interruption due to loss of electrical power, the laser will not restart until the operator presses the On button in the main control window in the computer interface (21 CFR 1040.10 (f)(10)).

Location of Safety Labels

Refer to Figures 1-1 through 1-11 for a description and location of all safety labels. These include warning labels indicating removable or displaceable protective housings, apertures through which laser radiation is emitted and labels of certification and identification [CFR 1040.10(g), CFR 1040.2, and CFR 1010.3/EN61010-1 EN60825-1, Clause 5].



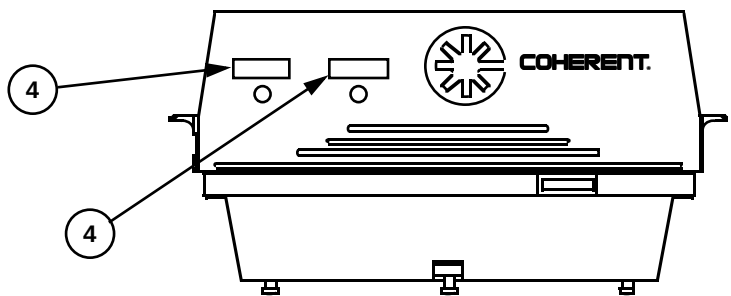
Laser Head Exterior, Top View



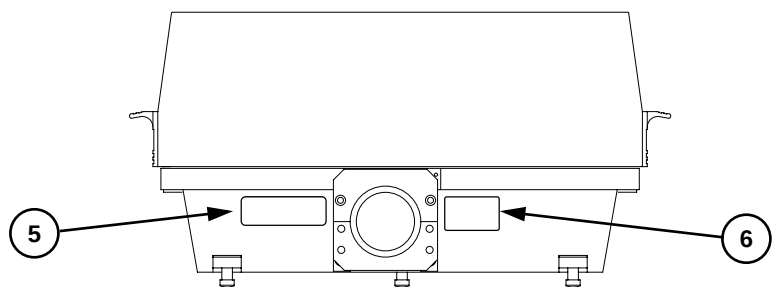
Laser Head Exterior, Side View

Note: Key is located on sheets 7 through 11.

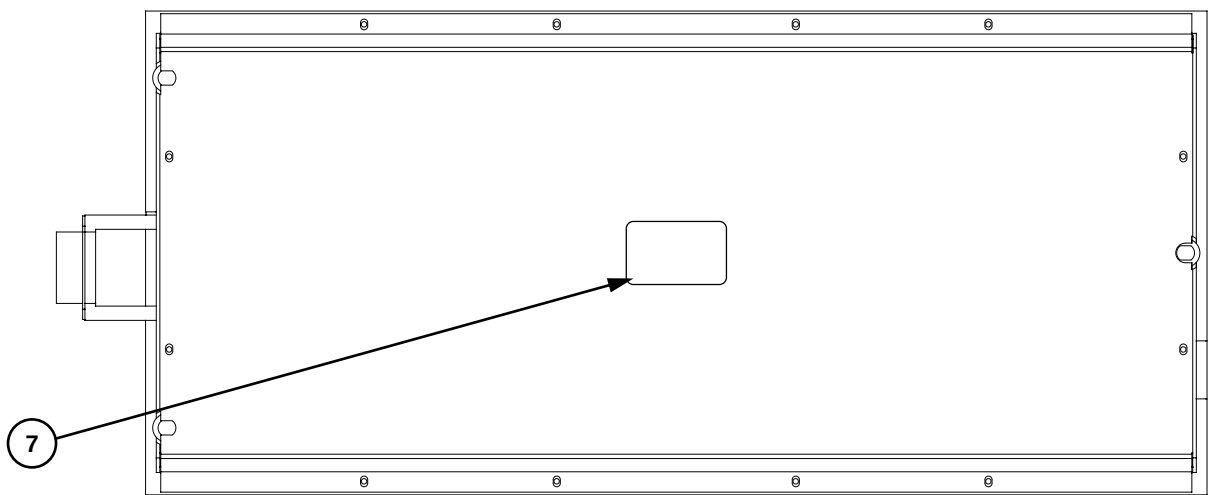
Figure 1-1. Location of Safety Labels (Sheet 1 of 11)



Laser Head Exterior, Front View

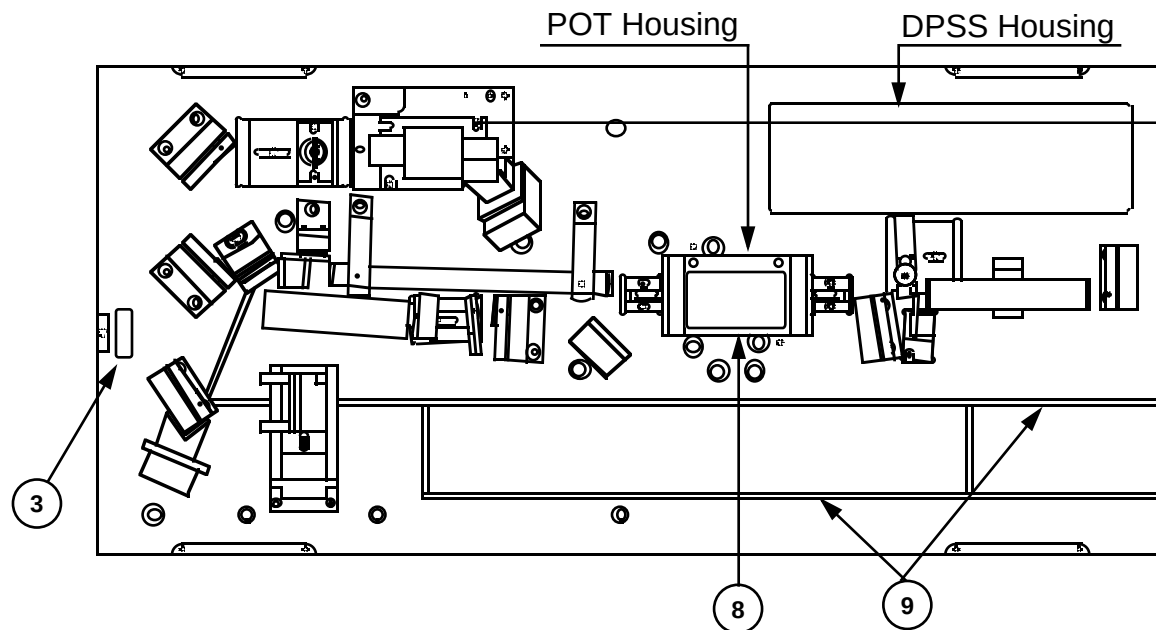


Laser Head Exterior, Rear View

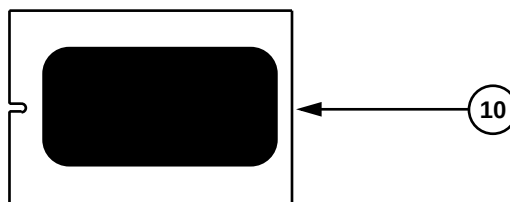


Laser Head Exterior, Bottom View

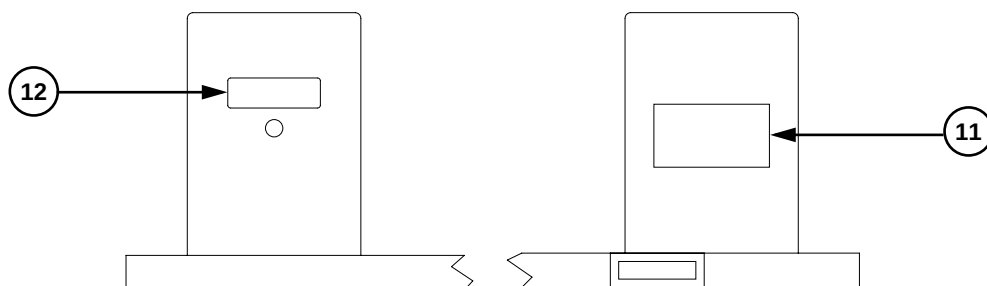
Note: Key is located on sheets 7 through 11.
Figure 1-2. Location of Safety Labels (Sheet 2 of 11)



Laser Head Interior Without Cover, Top View



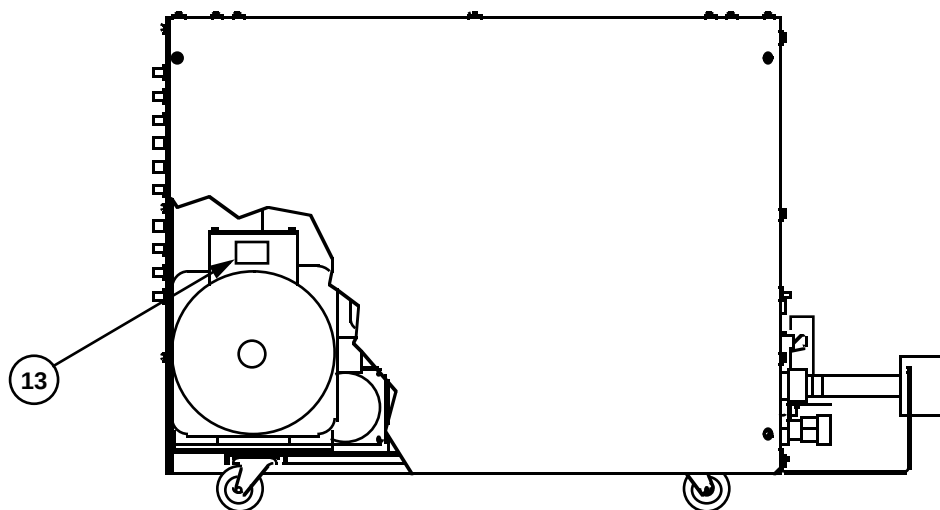
Laser Head Interior, POT Terminal Housing Cover



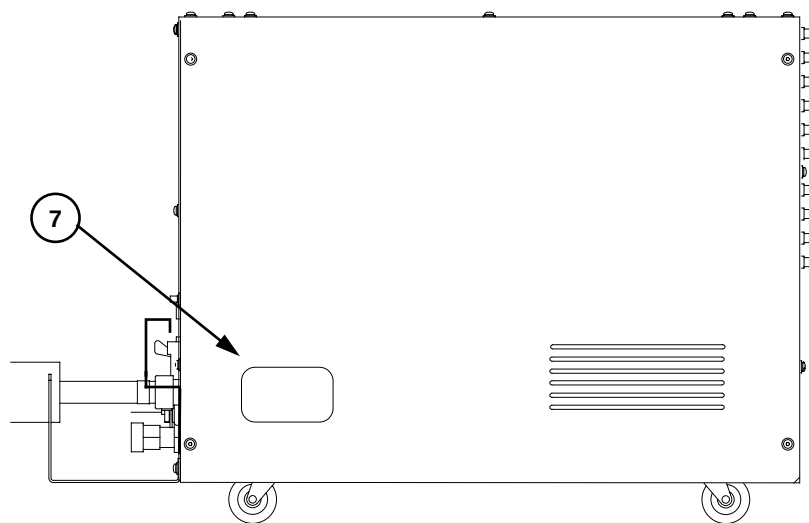
Laser Head Interior, DPSS Housing, Exit Aperture (12) and Rear (11)

Note: Key is located on sheets 7 through 11.

Figure 1-3. Location of Safety Labels (Sheet 3 of 11)



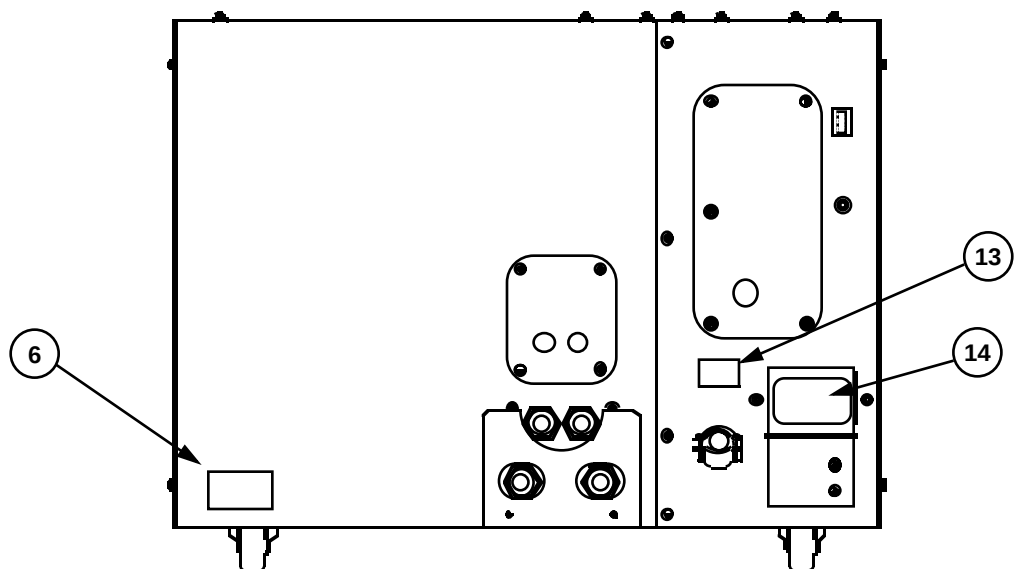
Power Supply, Cut-away, Right Side View, Water Pump



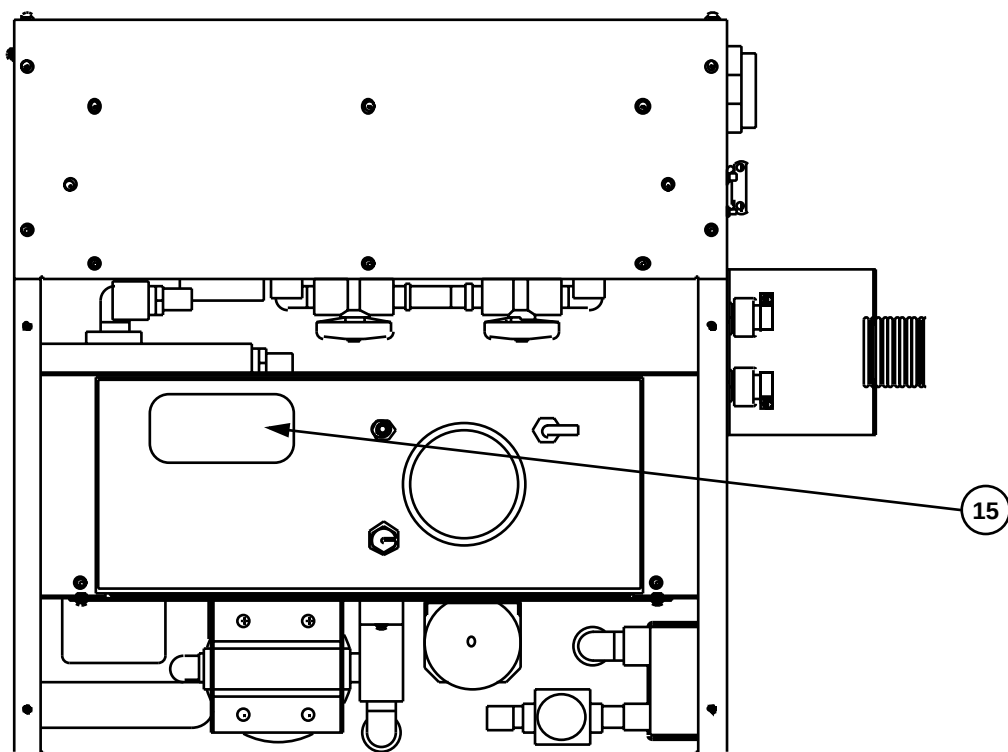
Power Supply, Left Side View

Note: Key is located on sheets 7 through 11.

Figure 1-4. Location of Safety Labels (Sheet 4 of 11)



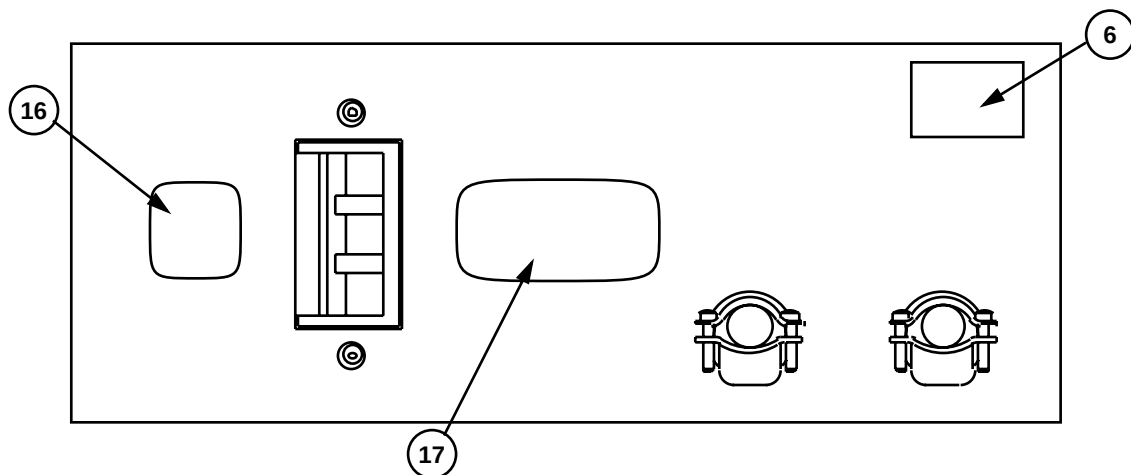
Power Supply, Rear View



Power Supply, Top View

Note: Key is located on sheets 7 through 11.

Figure 1-5. Location of Safety Labels (Sheet 5 of 11)



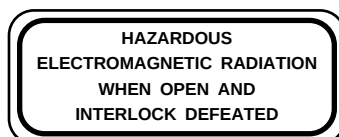
External EMI Filter (European Systems Only), Back View

Note: Key is located on sheets 7 through 11.

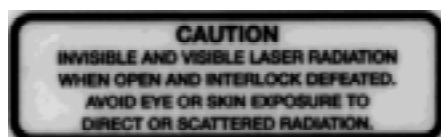
Figure 1-6. Location of Safety Labels (Sheet 6 of 11)



Label 1



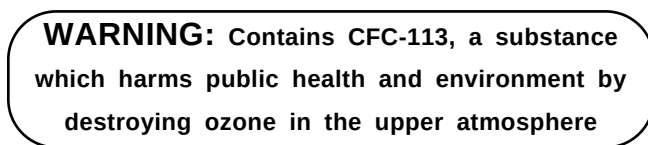
Label 2



Label 3



Label 4

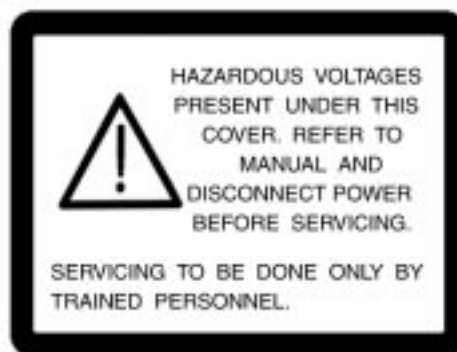


Label 5

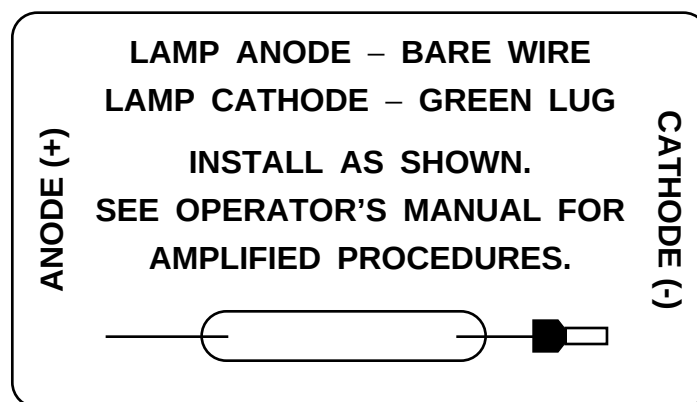
Figure 1-7. Location of Safety Labels (Sheet 7 of 11)



Label 6



Label 7

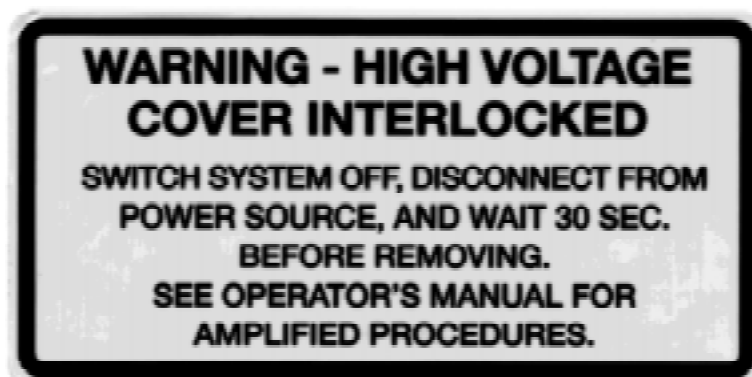


Label 8

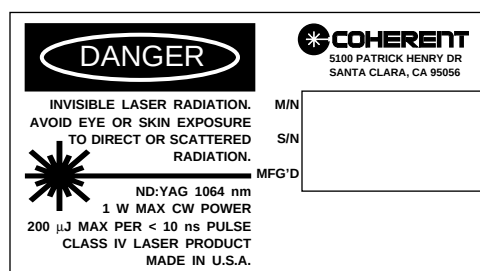
Figure 1-8. Location of Safety Labels (Sheet 8 of 11)



Label 9



Label 10



Label 11

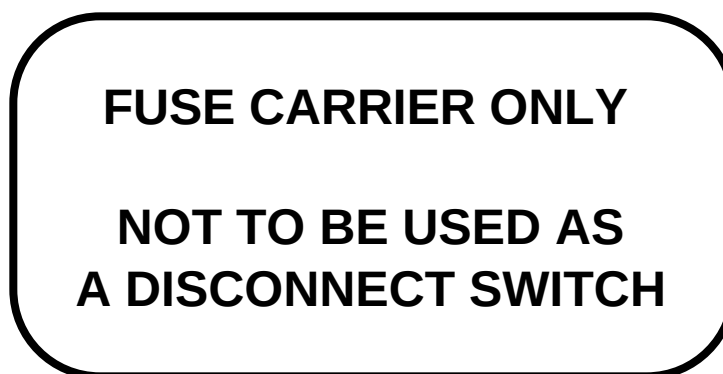


Label 12

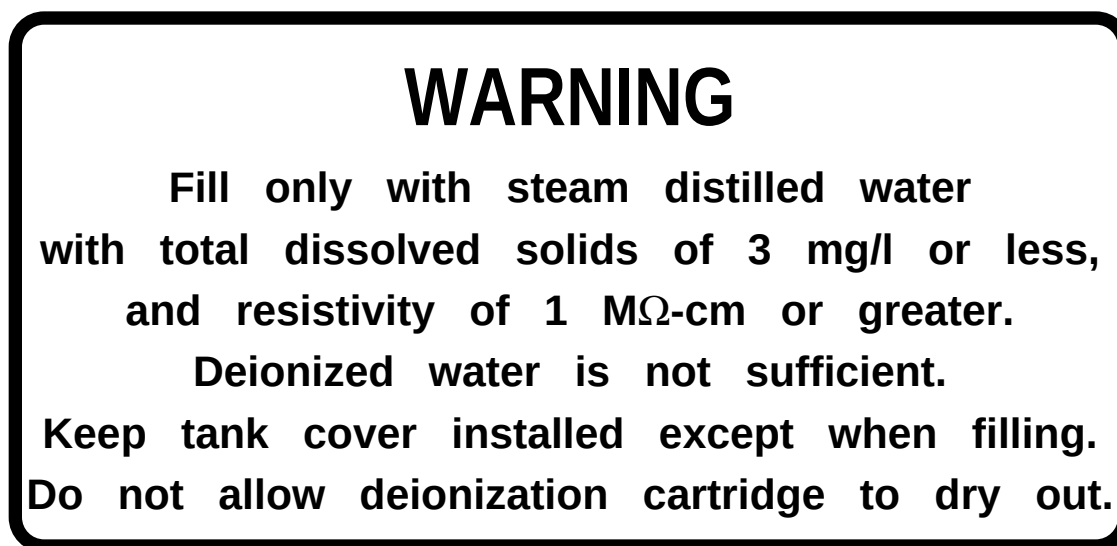
Figure 1-9. Location of Safety Labels (Sheet 9 of 11)



Label 13

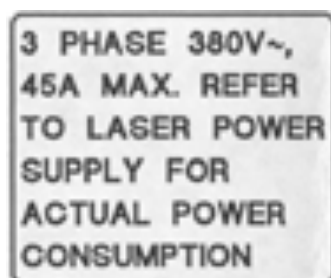


Label 14

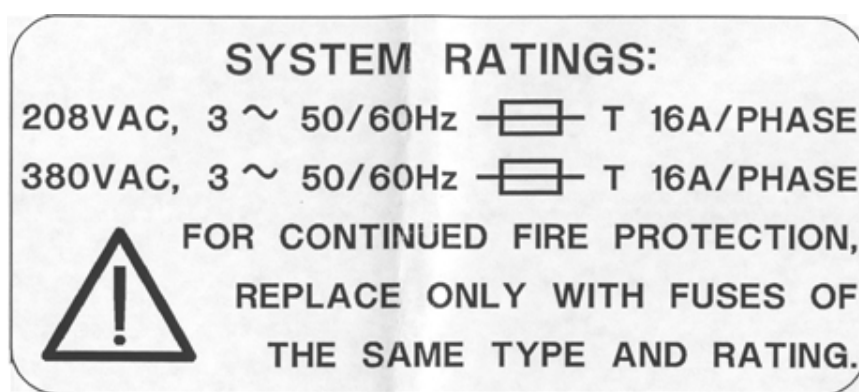


Label 15

Figure 1-10. Location of Safety Labels (Sheet 10 of 11)



Label 16



Label 17

Figure 1-11. Location of Safety Labels (Sheet 11 of 11)

References

More detailed information on laser safety can be found in the following publications:

"LIA Laser Safety Guide"

Laser Institute of America, 12424 Research Parkway, Suite 125, Orlando, FL 32826-3274, Telephone: (800)-345-2737 or (407)-380-1553

"American National Standard for the Safe Use of Lasers"

American National Standards Institute, 1430 Broadway, New York, NY 10018

"Safety with Lasers and Other Optical Sources"

D. Sliney and M. Wolbarsht, Plenum Publishing Company, New York, N.Y. 1980

Additional safety information and material can also be found through the following sources:

Laser Institute of America

Books, Safety standards, Training courses, Warning signs.
12424 Research parkway, Suite 125
Orlando, FL 32826-3274
Telephone: (800)-345-2737 or (407)-380-5588

Coherent, Inc.

"Laser Safety Comes to Light", Video

5100 Patrick Henry Drive
P.O. Box 54980
Santa Clara, CA 95056
Telephone: (408)-764-4800 or (800)-527-3786

Laser Focus World Buyer's Guide

List of suppliers of safety materials, e.g. signs, goggles etc.
One Technology Park Drive
P.O. Box 989
Westford, MA 01886
Telephone: (508)-692-0700

